Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An electrolytic cell contains a solution of Ag_2SO_4 and has platinum electrodes. A current is passed until $1.6gm$ of O_2 has been liberated at anode. The amount of silver deposited at cathode would be

(A) 107.88 g

(B) 1.6 g

(C) 0.8 g

(D) 21.60 g

Q.2

For the cell reaction, $Cu^{2+}\left(C_1, aq\right)+Zn(s)=Zn^{2+}\left(C_2, aq\right)+Cu(s)$ of an electrochemical cell, the change in free energy, ΔG, at a given temperature is a function of

(A) $\ln\left(C_1\right)$

(B) $\ln\left(C_2 / C_1\right)$

(C) $\ln\left(C_2\right)$

(D) $\ln\left(C_1+C_2\right)$

Q.3

The electrode potential $E_{\left(Zn^{2+}/Zn\right)}$ of a zinc electrode at $25^{\circ}C$ with an aqueous solution of $0.1MZnSO_4$ is $\left[E^{\circ}_{\left(Zn^{2+}/Zn\right)}=-0.76\ V.\right.$ Assume $\left.\frac{2.303RT}{F}=0.06 at 298\ K\right]$

(A) $+0.73$

(B) -0.79

(C) -0.82

(D) -0.70

Q.4

A gas X at $1\ atm$ is bubbled through a solution containing a mixture of $1MY^{-}$ and MZ^{-} at $25^{\circ}C$. If the reduction potential of $Z>Y>X$, then,

(A) Y will oxidize X and not Z

(B) Y will oxidize Z and not X

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(C) Y will oxidize both X and Z

(D) Y will reduce both X and Z

Q.5

For the electrochemical cell, $M|M^{+}||X^{-}|X$,
$E^{0}_{M^{+}/M} = 0.44V$ and $E^{o}_{(X/X^{-})} = 0.33\ V$
From this data one can deduce that

(A) $M + X \rightarrow M^{+} + X^{-}$ is the spontaneous reaction

(B) $M^{+} + X^{-} \rightarrow M + X$ is the spontaneous reaction

(C) $E_{cell} = 0.77\ V$

(D) $E_{cell} = -0.77\ V$

Q.6

Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration.
Some half cell reactions and their standard potentials are given below:
$MnO_4^-(aq.) + 8H^+(aq.) + 5e^- \rightarrow Mn^{2+}(aq.) + 4H_2O(l)$
$E^\circ = 1.51\ V$
$Cr_2O_7^{2-}\ (aq.) + 14H^+\ (aq.) + 6e^- \rightarrow 2Cr^{3+}\ (aq.) + 7H_2O(l)$
$E^\circ = 1.38\ V$
$Fe^{3+}\ (aq.) + e^- \rightarrow Fe^{2+}\ (aq.)$
$E^\circ = 0.77\ V$
$Cl_2(\ g) + 2e^- \rightarrow 2Cl^-\ (aq.)$
$E^\circ = 1.40\ V$
Identify the only incorrect statement regarding the quantitative estimation of aqueous $Fe(NO_3)_2$

(A) MnO_4^- can be used in aqueous HCl

(B) $Cr_2O_7^{2-}$ can be used in aqueous HCl

(C) MnO_4^- can be used in aqueous H_2SO_4

(D) $Cr_2O_7^{2-}$ can be used in aqueous H_2SO_4

Q.7

On the basis of the following E° values, the strongest oxidizing agent is :
$[Fe(CN)_6]^{4-} \rightarrow [Fe(CN)_6]^{3-} + e^-; E^\circ = -0.35\ V$
$Fe^{2+} \rightarrow Fe^{3+} + e^-; \quad E^\circ = -0.77\ V$

(A) $[Fe(CN)_6]^4$

(B) Fe^{2+}

(C) Fe^{3+}

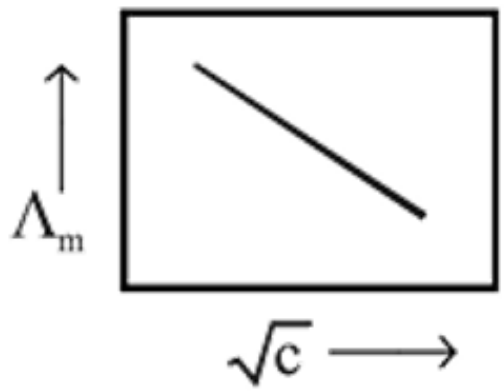
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(D) $[Fe(CN)_6]^{3-}$

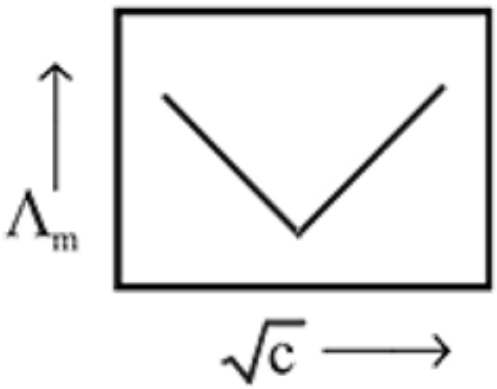
Q.8

The variation of molar conductance of strong electrolyte with (concentration) $^{1/2}$ is represented by

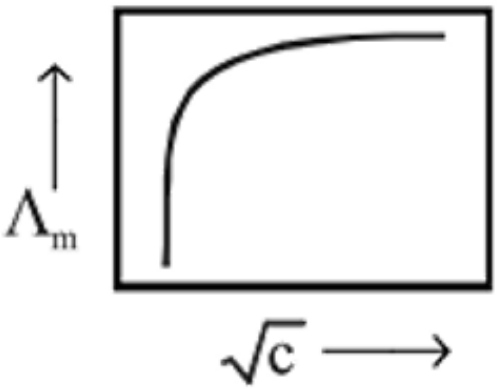
(A)



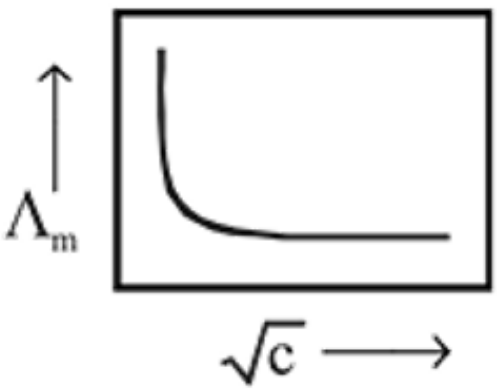
(B)



(C)



(D)



Q.9

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In acidic medium MnO_2 is an oxidant as $MnO_2(s) + 4H^+ + 2e^- \longrightarrow Mn^{2+} + 2H_2O$
If the pH of solution is decreased by one unit, the electrode potential of the half cell $Pt : MnO_2, Mn^{2+}$ will change by

(A) $0.236\ V$

(B) $-0.236\ V$

(C) $-0.118\ V$

(D) $0.118\ V$

Q.10

The standard EMF of Daniell cell is 1.10 volt. The maximum electrical work (in kJ) obtained from the Daniell cell is

(A) 213.2

(B) 211.3

(C) 212.3

(D) 212.2

Q.11

Standard free energies of formation (in kJ/mol) at $298\ K$ are $-237.2, -394.4$ and -8.2 for $H_2O(l), CO_2(g)$ and pentane (g) respectively. The value E°_{cell} for the pentane-oxygen fuel cell is Answer upto 3 decimal places without rounding off.

(A) 1.906

(B) 1.960

(C) 1.069

(D) 1.096

Q.12

Kohlrausch's law states that at:

(A) finite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte.

(B) infinite dilution each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of the electrolyte.

(C) infinite dilution, each ion makes definite contribution to conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(D) infinite dilution, each ion makes definite contriubtion to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte.

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Q.13

The standard emf for the given cell reaction, $Zn + Cu^{2+} \rightarrow Cu + Zn^{2+}$ is 1.10 V at 25°C. The emf for the cell reaction, when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solutions are used, at 25°C, is
(Use $\frac{2.303RT}{F} = 0.059$)

(A) -1.10 V

(B) 2.20 V

(C) -2.20 V

(D) 1.10 V

Q.14

The standard reduction potentials at 298 K for the following half reactions are given against each
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s); -0.762$ V
$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s); -0.740$ V
$2H^+(aq) + 2e^- \rightleftharpoons H_2(g); 0.00$ V
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq); 0.770$ V
Which is the strongest reducing agent?

(A) $Zn(s)$

(B) $Cr(s)$

(C) $H_2(g)$

(D) $Fe^{3+}(aq)$

Q.15

Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because

(A) Zn acts as oxidizing when reacts with HNO_3

(B) HNO_3 is weaker acid than H_2SO_4 and HCl.

(C) In electrochemical series Zn is above hydrogen.

(D) NO_3^- is reduced in preference to hydronium.

Q.16

A smuggler could not carry gold by depositing iron on the gold surface since

(A) gold is denser

(B) iron rusts

(C) gold has higher reduction potential than iron

(D) gold has lower reduction potential than iron

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Q.17

Standard electrode potential for Sn^{4+}/Sn^{2+} couple is $+0.15\ V$ and that for the Cr^{3+}/Cr couple is $-0.74\ V$. These two couples in their standard state are connected to make a cell. The cell potential will be:

(A) $+1.19\ V$

(B) $+0.89\ V$

(C) $+0.18\ V$

(D) $+1.83\ V$

Q.18

A solution contains Fe^{2+}, Fe^{3+} and I^- ions. This solution was treated with iodine at $35°C$. $E°$ for Fe^{3+}/Fe^{2+} is $+0.77\ V$ and $E°$ for $I_2/2I^- = 0.536\ V$

The favourable redox reaction is :

(A) I_2 will be reduced to I^-

(B) There will be no redox reaction

(C) I^- will be oxidised to I_2

(D) Fe^{2+} will be oxidised to Fe^{3+}

Q.19

Consider the following four electrodes:

$P = Cu^{2+}(0.0001M)/Cu(s)$

$Q = Cu^{2+}(0.1M)/Cu(s)$

$R = Cu^{2+}(0.01M)/Cu(s)$

$S = Cu^{2+}(0.001M)/Cu(s)$

If the standard reduction potential of Cu^{2+}/Cu is $+0.34\ V$, the reduction potentials in volts of the above electrodes follow the order.

(A) $P > S > R > Q$

(B) $S > R > Q > P$

(C) $R > S > Q > P$

(D) $Q > R > S > P$

Q.20

Following cell has EMF $0.7995\ V$.

$Pt\,|H_2(1\ atm)|\,HNO_3(1M)||AgNO_3(1M)\ |\ Ag$

If we add enough KCl to the Ag cell so that the final Cl^- is $1M$. Now the measured emf of the cell is $0.222\ V$. The K_{sp} of $AgCl$ would be-

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(A) $1 \times 10^{-9.8}$

(B) $1 \times 10^{-19.6}$

(C) 2×10^{-10}

(D) 2.64×10^{-14}

Answers & Solutions

Q.1 Answer:

21.60 g

Solution:

$\frac{W_A}{E_A} = \frac{W_B}{E_B}; \frac{1.6}{8} = \frac{Wt.\text{ of } Ag}{108}$

$\therefore$ **Wt. of** $Ag = 21.6$ g

Q.2 Answer:

$\ln (C_2/C_1)$

Solution:

$\Delta G = -nE^{\circ}F$

For concentration cell, $E = \frac{RT}{nF} \ln \frac{C_2}{C_1}$

In it R, T, n **and** F **are constant**

So E **is based upon** $\ln C_2/C_1$

Now $\Delta G = -nEF = -nF \times \frac{RT}{nF} \ln C_2/C_1$

$= -RT \ln C_2/C_1$

At constant temperature ΔG **is based upon** $\ln (C_2/C_1)$.

Q.3 Answer:

-0.79

Solution:

$$Zn^{2+} \rightarrow Zn$$

For

$$E_{Zn^{2+}/Zn} = E^{\circ}_{Zn^{2+}/Zn} - \frac{2.303RT}{nF} \log \frac{[Zn]}{[Zn^{2+}]}$$

$$= -0.76 - \frac{0.06}{2} \log \frac{1}{[0.1]} = -0.76 - 0.03$$

$$E_{Zn^{2+}/Zn} = -0.79 \text{ V}$$

Q.4 Answer:

Y will oxidize X and not Z

Solution:

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The given order of reduction potentials (or tendencies) is $Z > Y > X$. A spontaneous reaction will have the following characteristics Z reduced and Y oxidised Z reduced and X oxidised Y reduced and X oxidised Hence, Y will oxidise X and not Z.

Q.5 Answer:

$\mathrm{M^+ + X^- \rightarrow M + X}$ is the spontaneous reaction

Solution:

For, $\mathrm{M^+ + X^- \longrightarrow M + X, E^o_{cell} = 0.44 - 0.33 = 0.11\ V}$
is positive, hence reaction is spontaneous.

Q.6 Answer:

$\mathrm{MnO_4^-}$ can be used in aqueous HCl

Solution:

$\mathrm{MnO_4^-}$ **will oxidise** $\mathrm{Cl^-}$ **ion according to the equation**

$$\mathrm{2MnO_4^- + 16H^+ + 10Cl^- \longrightarrow 2Mn^{2+} + 8H_2O + 5Cl_2 \uparrow}$$

The cell corresponding to this reaction is as follows:

$$\mathrm{Pt, Cl_2(1\ atm) \left| Cl^- \right\| MnO_4^-, Mn^{2+}, H^+ \right| Pt}$$

$$\mathrm{E^o_{cell} = 1.51 - 1.40 = 0.11\ V}$$

$\mathrm{E^o_{cell}}$ **being** $+\mathrm{ve}$, $\Delta \mathrm{G^\circ}$ **will be -ve and hence the above reaction is feasible.** $\mathrm{MnO_4^-}$ **will not only oxidise** $\mathrm{Fe^{2+}}$ **ion but also** $\mathrm{Cl^-}$ **ion simultaneously.**

Q.7 Answer:

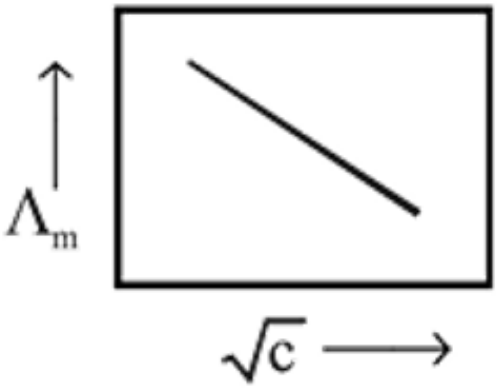
$\mathrm{Fe^{3+}}$

Solution:

From the given data we find $\mathrm{Fe^{3+}}$ **is strongest oxidising agent. More the positive value of** $\mathrm{E^\circ}$**, more is the tendency to get oxidized. Thus correct option is** (c).

Q.8 Answer:

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Solution:

In case of molar conductance of strong electrolyte there is little increase with dilution.

Q.9 Answer:

$0.118\ \mathrm{V}$

Solution:

$$\mathrm{E} = \mathrm{E}^{\circ} - \frac{0.0592}{2} \log \frac{\left[\mathrm{Mn}^{2+}\right]}{\left[\mathrm{H}^{+}\right]^{4}}$$

$$= \mathrm{E}^{\circ} - \frac{0.0592 \times 4}{2} \log \frac{\left[\mathrm{Mn}^{2+}\right]^{1/4}}{\left[\mathrm{H}^{+}\right]}$$

$$= \mathrm{E}^{\circ} - 0.0592 \times 2 \left(\log \left[\mathrm{Mn}^{2+}\right]^{1/4} + \mathrm{pH}\right)$$

$$\Delta \mathrm{E} = \mathrm{E}_2 - \mathrm{E}_1 = 0.0592 \times 2 \left(\mathrm{pH}_1 - \mathrm{pH}_2\right)$$

$$= 0.118 \times 1 = 0.118\ \mathrm{V}$$

Q.10 Answer:

212.3

Solution:

$$\Delta \mathrm{G} = -\mathrm{nF}\mathrm{E}^{\circ}{}_{\text{cell}} = -2 \times 96500 \times 1.1\ \mathrm{J} = 212.3\ \mathrm{kJ}.$$

Q.11 Answer:

1.096

Solution:

Writing the equation for pentane-oxygen fuel cell at respective electrodes and overall reaction, we get

At Anode:

$$\mathrm{C_5H_{12}} + 10\mathrm{H_2O} \rightarrow 5\mathrm{CO_2} + 32\mathrm{H^+} + 32\mathrm{e^-}$$

(pentane) At Cathode:

$$\frac{8\mathrm{O_2} + 32\mathrm{H^+} + 32\mathrm{e^-} \rightarrow 16\mathrm{H_2O}}{\text{Overall}: \mathrm{C_5H_{12}} + 8\mathrm{O_2} \rightarrow 5\mathrm{CO_2} + 6\mathrm{H_2O}}$$

Calculation of ΔG° for the above reaction $\Delta G^{\circ} = [5 \times (-394.4) + 6 \times (-237.2)] - [-8.2]$

$= -1972.0 - 1423.2 + 8.2 = -3387.0\ \mathrm{kJ}$

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$= -3387000$ **Joules. From the equation we find** $\mathrm{n} = 32$ **Using the relation,** $\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}}$ **and substituting**

various values, we get

$-3387000 = -32 \times 96500 \times E^{\circ}_{\text{cell}}\ (\mathrm{F} = 96500\mathrm{C})$

or $E^{\circ}_{\text{cell}} = \frac{3387000}{32 \times 96500}$

$= \frac{3387000}{3088000}$ **or** $\frac{3387}{3088}\ \mathrm{V} = 1.0968\ \mathrm{V}$

Q.12 Answer:

infinite dilution, each ion makes definite contriubtion to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte.

Solution:

Kohlrausch s Law states that at infinite dilution, each ion migrates independently of its co-ion and contributes to the total equivalent conductance of an electrolyte a definite share which depends only on its own nature. From this definition we can see that option (d) **is the correct answer.**

Q.13 Answer:

$1.10\ \ \mathrm{V}$

Solution:

For the reaction:

$\mathrm{Zn} + \mathrm{Cu}^{2+} \rightarrow \mathrm{Cu} + \mathrm{Zn}^{2+}$

$\mathrm{E_{cell}} = \mathrm{E^{\circ}_{cell}} - \frac{0.059}{\mathrm{n}} \log \frac{\mathrm{Zn}^{2+}}{\mathrm{Cu}^{2+}}$

$= 1.10 - \frac{0.059}{2} \log \frac{0.1}{0.1}$

$= 1.10\ \ \mathrm{V}$

Q.14 Answer:

$\mathrm{Zn}(s)$

Solution:

Since oxidation potential of Zn **is highest hence strongest reducing agent.**

Q.15 Answer:

$\mathrm{NO_3^-}$ is reduced in preference to hydronium.

Solution:

The reduction potential of $\mathrm{NO_3^-}$ **ion is more than** $\mathrm{H^+}$ **ion.**

Q.16 Answer:

gold has higher reduction potential than iron

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Solution:

Gold having higher E^0_{Red} and oxidises $\text{Fe} \longrightarrow \text{Fe}^{2+}$
Gold has a reduction potential $= +1.5$ V.
Iron has a reduction potential $= -0.44$ V.
Thus, gold has a higher reduction potential than iron.
An element with a higher reduction potential is a stronger oxidizing agent.
So, gold is a stronger oxidizing agent than iron.
Hence, gold will get reduced itself instead of iron.
Therefore, iron can not be deposited on the gold surface.

Q.17 Answer:

$+0.89$ V

Solution:

Given $\text{E}_{\text{Sn}^{4+}/\text{Sn}^{2+}} = +0.15$ V
$\text{E}_{\text{Cr}^{3+}}/\text{Cr} = -0.74$ V
$E^{\text{o}}_{\text{cell}} = \text{E}^{\text{o}}_{\text{cathode}} - \text{E}^{\text{o}}_{\text{anode}}$
$= 0.15 - (-0.74) = +0.89$ V

Q.18 Answer:

I^- will be oxidised to I_2

Solution:

Given $\text{Fe}^{3+}/\text{Fe}^{2+} = +0.77$ V
and $\text{I}_2/2\text{I}^- = 0.536$ V
$2\left(\text{e}^- + \text{Fe}^{3+} \longrightarrow \text{Fe}^{2+}\right) E^\circ = 0.77$ V
$2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$
$E^\circ = -0.536$ V
$2\text{Fe}^{3+} + 2\text{I}^- \longrightarrow 2\text{Fe}^{2+} + \text{I}_2$
$E^\circ = E^\circ_{\text{ox}} + E^\circ{}_{\text{red}}$
$= 0.77 - 0.536 = 0.164$ V
So, reaction will take place.

Q.19 Answer:

$\text{Q} > \text{R} > \text{S} > \text{P}$

Solution:

$\text{E}_{\text{red}} = \text{E}^{\text{o}}_{\text{red}} + \frac{0.591}{\text{n}} \log [\text{M}^{\text{n}+}]$
Lower the concentration of $\text{M}^{\text{n}+}$, lower is the reduction potential. Hence order of reduction potential is :
$\text{Q} > \text{R} > \text{S} > \text{P}$

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Q.20 Answer:

$1 \times 10^{-9.8}$

Solution:

$2\mathrm{Ag}^{+} + \mathrm{H}_2 \longrightarrow 2\mathrm{H}^{+} + 2\mathrm{Ag}$

$E = E^{\circ} - \frac{0.0591}{2} \log \frac{\left[\mathrm{H}^{+}\right]^2}{\mathrm{P}_{\mathrm{H}_2} \times \left[\mathrm{Ag}^{+}\right]^2}$

$0.222 = 0.7995 - \frac{0.0591}{2} \log \frac{1}{\left[\mathrm{Ag}^{+}\right]^2}$

$\left[\mathrm{Ag}^{+}\right] = 10^{-9.8}$

$K_{sp} = \left[\mathrm{Ag}^{+}\right]\left[\mathrm{Cl}^{-}\right] = \left(10^{-9.8}\right) \times (1) = 10^{-9.8}$

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